

tmfast: Fast Topic Models via Varimax-Rotated PCA

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Software

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Summary

Topic modeling is a widely used natural language processing technique for discovering latent thematic structure in large text corpora. Most topic modeling approaches — including Latent Dirichlet Allocation (LDA) — rely on Bayesian inference, which is computationally intensive, stochastic, and introduces researcher degrees of freedom through the choice of prior distributions.

tmfast is an R package that implements a fast, deterministic alternative using varimax-rotated principal component analysis (PCA), following the “vintage factor analysis” approach of Rohe & Zeng (2020). The package leverages truncated PCA via the `irlba` package (Jim Baglama et al., 2022) to efficiently handle the sparse term-document matrices typical of text data, fitting models roughly an order of magnitude faster than the widely used `stm` package (Roberts et al., 2019). Because fitting is deterministic, results are fully reproducible across runs without setting a random seed. Additional features include an information-theoretic approach to vocabulary selection, broom-compatible tidiers for extracting word-topic and topic-document distributions into tidy data workflows, and simulation samplers for benchmarking and method evaluation.

Statement of Need

Topic modeling is widely used by computational social scientists, digital humanists, and researchers working with large text corpora to discover latent thematic structure (Blei et al., 2003; Roberts et al., 2019). A key advantage over other dimensionality reduction methods is that topic modeling simultaneously clusters both terms and documents, enabling analysts to assign human-meaningful, domain-specific labels to the discovered topics.

However, standard topic modeling packages are slow, stochastic, and introduce researcher degrees of freedom through the choice of Bayesian prior distributions. Because fitting even a single model is computationally expensive, analysts rarely fit and compare multiple models — arguably the most principled way to assess sensitivity of results to methodological choices (Gelman & Loken, 2013; Steegen et al., 2016). Instead, they typically settle on a single “best” model selected by informal assessments of topic interpretability, itself an additional source of researcher degrees of freedom.

tmfast addresses these problems directly. Its fitting algorithm is deterministic — results are fully reproducible without setting a random seed — and requires no specification of Bayesian priors. In tests with simulated data, tmfast fits topic models approximately 20× faster than `stm` (Roberts et al., 2019), making it practical to fit and compare many models as part of a routine analysis workflow.

State of the Field

Topic modeling in R is dominated by packages implementing Latent Dirichlet Allocation (LDA)-based approaches (Blei et al., 2003). The `topicmodels` package (Grün & Hornik, 2011)

40 provides LDA and correlated topic models via variational Bayes and Gibbs sampling. The
41 `stm` package (Roberts et al., 2019) extends LDA to incorporate document-level metadata as
42 covariates on topic prevalence and content, and is generally regarded as the state of the art for
43 social-scientific text analysis in R. Both packages rely on iterative Bayesian inference, which is
44 computationally intensive, stochastic (requiring random seeds for reproducibility), and sensitive
45 to prior specification.

46 Rohe & Zeng (2020) propose an alternative approach, approaching topic modeling using
47 principal component analysis (PCA) followed by a varimax rotation. They show (Lemma 5.2)
48 that the term-document occurrence rate matrix can be approximately factored into term-topic
49 and topic-document distributions — the same quantities estimated by LDA — and that PCA
50 with varimax rotation provides statistically consistent estimates of these latent distributions.
51 The varimax rotation promotes sparsity in the estimated factor loadings, giving topics the
52 interpretability property that makes them useful: each topic is characterized by a small number
53 of high-weight terms, and each document is associated with a small number of dominant
54 topics.

55 The PCA-based approach offers two practical advantages over LDA. First, it is deterministic
56 and fully reproducible without random seeds. Second, efficient partial SVD algorithms for
57 sparse matrices (James Baglama & Reichel, 2005) make it substantially faster than iterative
58 Bayesian methods — in a test case, `tmfast` fits topic models approximately 20× faster than
59 `stm`.

60 The topic modeling approach used in `tmfast` does have three notable limitations compared
61 to `stm`. First, there is no way to incorporate document-level metadata into the models.
62 Second, scores and loadings for varimax-rotated partial PCA are not directly interpretable as
63 probability distributions. Casting them into distributions introduces a notable researcher degree
64 of freedom, comparable to the choice of Bayesian priors of LDA. Third, in tests with simulated
65 data, `tmfast` was somewhat less accurate than `stm` at recovering true topic-document and
66 word-topic distributions.

67 Software Design

68 Fitting pipeline

69 `tmfast` fits topic models in two steps. First, `irlba::prcomp_irlba()` is used to compute a
70 truncated PCA of the document-term matrix, retaining only the top principal components
71 needed for the largest requested number of topics (k). Because PCA is computed only
72 once, multiple models with different values of k can be extracted from a single fit, applying
73 varimax rotation at each requested k without repeating the—relatively computationally
74 intensive—truncated PCA step.

75 Second, `stats::varimax()` is used to apply an orthogonal rotation to the top- k loading matrix
76 for each value of k . To ensure consistent sign conventions, factors with negative skew are
77 automatically reflected; this makes topics “peak” rather than “valley” shaped without affecting
78 any statistical properties.

79 The output is a `tmfast` object storing the PCA fit, varimax rotation matrices, and rotated
80 loadings and scores for each requested k .

81 Vocabulary selection

82 For large corpora, restricting the vocabulary before fitting substantially reduces memory use
83 and fitting time. `tmfast` provides two information-theoretic vocabulary selection metrics as
84 alternatives to TF-IDF.

85 `ndH()` measures the information gain a term carries beyond what would be expected if it were

distributed uniformly across documents. More precisely, it involves $\Delta H = \log_2(D) - H(p)$, equivalent to KL divergence from the uniform distribution, where D is the number of documents and $H(p)$ is the observed entropy of the term's document distribution. ΔH is then weighted by logged term frequency $\log_2(n)$ to down-weight rare terms that may be OCR artifacts or misspellings: $ndH = \log_2(n) \cdot \Delta H$.

`ndR()` uses information gain/KL divergence relative to a length-proportional document distribution rather than the uniform distribution: $p(d) = l_d / \sum_d l_d$ where l_d is the word length of document d , so that short documents (such as short articles) are less probable than long documents (such as books). We then calculate KL divergence for the word from this length-proportional distribution: $\Delta R = \sum_d p(d | w) \log_2[p(d | w) / p(d)]$. Because short documents are less probable, terms that are concentrated to short documents are “surprising” (high information gain) and thus tend to have a greater ΔR , making this approach preferable when document lengths vary significantly. We then weight by $\log_2(n)$ as with `ndH`.

In practice, analysts select a target vocabulary size and retain the top-ranked terms by either metric.

Tidiers and post-processing

`tmfast` implements a broom-compatible `tidy()` method that extracts word-topic (β) and topic-document (γ) distributions as long-format data frames, compatible with `tidyverse` (Wickham et al., 2019) workflows and the `tidytext` package (Silge & Robinson, 2016). The `tidy()` method requires specifying the value of k ; `tidy_all()` is provided as a convenience for extracting all values of k included in the `tmfast` object in a single line.

Because varimax loadings can be negative and generally do not sum to one, the tidiers apply a softmax transformation: $\beta_{wt} = \exp(L_{wt}) / \sum_w \exp(L_{wt})$ for word w and topic t , converting raw loadings into proper probability distributions. The same transformation is applied to document scores for γ .

In development, it was found that these softmax-normalized distributions tend to be flatter than the Dirichlet distributions that LDA assumes. To address this, `tidy()` accepts an optional exponent argument that applies a power renormalization: $\beta'_{wt} = \beta_{wt}^e / \sum_w \beta_{wt}^e$. Higher exponents sharpen distributions toward fewer dominant terms. `target_power()` automates the choice of exponent by finding the value whose output matches a researcher-specified target entropy. This target can be computed using `expected_entropy()`, which returns the expected Shannon entropy of a symmetric Dirichlet given concentration parameter α . This replaces Bayesian prior specification in LDA, with a similar effect: the analyst chooses how concentrated or spread they want topics and documents to be. This also introduces an important research degree of freedom.

Simulation samplers

`tmfast` includes a suite of simulation tools for benchmarking and method evaluation. `rdirichlet()` samples from the Dirichlet distribution with symmetric or asymmetric concentration parameters. `peak_alpha()` constructs an asymmetric concentration vector with a single dominant component, useful for generating “pure” documents that belong primarily to one topic. `draw_corpus()` implements the standard LDA generative process: for each document, it draws a topic mixture $\theta_d \sim \text{Dirichlet}(\alpha)$, then draws each word from the corresponding topic-word distribution ϕ_k . When the `furrr` package is available, corpus generation is parallelized automatically.

`journal_specific()` provides a complete benchmarking simulation: it generates a corpus with known true topic and word distributions (“specific” to each simulated “journal”), fits `tmfast`, aligns fitted topics to true topics as a linear assignment problem, and returns recovery accuracy as mean Hellinger distances for both ϕ (true word-topic) and θ (true topic-document)

distributions. This enables systematic evaluation of how recovery accuracy scales with corpus size, vocabulary size, and number of topics.

Research Impact Statement

tmfast was originally developed to support the text analysis in Hicks & Lobato (2025), which used topic models to study race science in mainstream psychology journals from 1960 to 2010. tmfast has subsequently been developed as a standalone package for the broader text mining community. The package has an arXiv preprint (arXiv:2305.01535, posted May 2023) and was first released on CRAN (version 0.1.1) in May 2026.

AI Usage Disclosure

Claude Sonnet 4.6, via Claude Code, was used to prepare this paper. Claude was also used to prepare tmfast for CRAN submission, especially writing unit tests and documentation. Prior to March 2026 no generative AI was used to develop the package. I (the human author) confirm that I reviewed, edited, and validated all AI-assisted outputs and made the core design decisions.

References

- Baglama, James, & Reichel, L. (2005). Augmented implicitly restarted lanczos bidiagonalization methods. *SIAM Journal on Scientific Computing*, 27(1), 19–42. <https://doi.org/10.1137/04060593X>
- Baglama, Jim, Reichel, L., & Lewis, B. W. (2022). *Irlba: Fast truncated singular value decomposition and principal components analysis for large dense and sparse matrices* (Version 2.3.5.1). <https://cran.r-project.org/web/packages/irlba/index.html>
- Blei, D. M., Ng, A. Y., & Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine Learning Research*, 3, 993–1022. <http://dl.acm.org/citation.cfm?id=944937>
- Gelman, A., & Loken, E. (2013). *The garden of forking paths: Why multiple comparisons can be a problem, even when there is no “fishing expedition” or “p-hacking” and the research hypothesis was posited ahead of time*. http://www.stat.columbia.edu/~gelman/research/unpublished/p_hacking.pdf
- Grün, B., & Hornik, K. (2011). Topicmodels: An R package for fitting topic models. *Journal of Statistical Software*, 40(13), 1–30. <https://doi.org/10.18637/jss.v040.i13>
- Hicks, D. J., & Lobato, E. J. C. (2025). Race science in mainstream psychology, 1960–2010. *Quantitative Science Studies*, 6, 879–900. <https://doi.org/10.1162/qss.a.11>
- Roberts, M. E., Stewart, B. M., & Tingley, D. (2019). Stm: An r package for structural topic models. *Journal of Statistical Software*, 91, 1–40. <https://doi.org/10.18637/jss.v091.i02>
- Rohe, K., & Zeng, M. (2020). *Vintage factor analysis with varimax performs statistical inference*. arXiv. <https://doi.org/10.48550/arXiv.2004.05387>
- Silge, J., & Robinson, D. (2016). Tidytext: Text mining and analysis using tidy data principles in R. *The Journal of Open Source Software*, 1(3), 37. <https://doi.org/10.21105/joss.00037>
- Steege, S., Tuerlinckx, F., Gelman, A., & Vanpaemel, W. (2016). Increasing transparency through a multiverse analysis. *Perspectives on Psychological Science*, 11(5), 702–712. <https://doi.org/10.1177/1745691616658637>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemond, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M.,

176 Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., ... Yutani, H. (2019). Welcome
177 to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. [https://doi.org/10.21105/](https://doi.org/10.21105/joss.01686)
178 [joss.01686](https://doi.org/10.21105/joss.01686)

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